



Session 100%

Assignment 2/2

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Help Requests

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Your Score: 100 Max Score: 100

Q 1

Q 1

Q 2

Q 3

Q 4

Q2. Bit Compression

Solved



Stuck somewhere?

Ask for help from a TA and get it resolved.

Get help from TA.

Problem Description

Richard Hendricks, a mastermind in compression algorithms, is an employee of Hooli in Silicon Valley. One day, he finally decided to quit and work on his new idea of the middle-out compression algorithm.

He needed to work at the bit - level to compress data. He, eventually, encountered this problem.

There is an array **A** of **N** integers. He has to perform certain operations on the elements.

In any one operation, two indices **i** and **j** ($i < j$) are chosen, and **A[i]** is replaced with **A[i] & A[j]**,

and **A[j]** is replaced with **A[i] | A[j]**, where **&** represents the Bitwise AND operation and **|** represents the Bitwise OR operation.

This operation is performed over all the pairs of integers in the array.

Help Richard find the Bitwise XOR of all the elements after performing the operations.

Problem Constraints

 $1 \leq N \leq 10^5$ $1 \leq A_i \leq 10^9$

Input Format

The first argument is the integer array A.

Output Format

Return a single integer denoting the XOR of the elements after performing the operations.

Example Input

Input 1:

A = [1, 3, 5]

Input 2:

A = [1, 1, 1]

Example Output

Output 1:

7

Output 2:

1

Example Explanation

Explanation 1:

After performing the operations, the array becomes [1, 1, 7].

The XOR of all the elements of this array is 7.

There can be other possible arrays, but we can prove that the XOR will always be 7.

Explanation 2:

Since all the elements are equal.

Even after applying the operations in any order, the array will be [1, 1, 1].

The XOR of all the elements of this array is 1.

See Expected Output

Java 8 (Array Support)

```
1 import java.util.Scanner;
2
3 public class Solution
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         String[] input = sc.nextLine().split(" ");
9         int[] A = new int[input.length];
10
11         for(int i = 0; i < input.length; i++)
12         {
13             A[i] = Integer.parseInt(input[i]);
14         }
15     }
16 }
```

Test Output Code's All Neat!

Compiling your Code...

> Success!

Running Test Cases...

> Test Case - Easy Success

> Test Case - Hard Success

Final Verdict

Test

Test With Custom Input

Submit